PRATHAM KUMAR

AI / GenAI Engineer — LLM Evaluation, Agentic Systems, RAG
Delhi, India | +91 8076123862 | prathamkumars125@gmail.com | github.com/prathamkumar125 | linkedin.com/in/pratham-kumar125

SUMMARY

AI Engineer with **1+ year at Google (via Highspring)** and **3 prior AI/ML internships**. Built **LLM evaluation frameworks (1,000+ responses/day, ~85% F1, ~90% manual QA automated)**, ML + LLM pipelines across **~5M entities**, and RAG/agentic systems in LangChain and LangGraph.

TECHNICAL SKILLS

Languages & Data: Python, SQL, MongoDB, C/C++, Power BI
AI/ML: PyTorch, TensorFlow, Scikit-learn, Transformers, OpenCV, NumPy, Pandas
Agentic & GenAI: LangChain, LangGraph, RAG, LLM Evaluation, LangSmith, Prompt Engineering, Hugging Face, MCP, N8N
Development & Cloud: FastAPI, Flask, REST API, Docker, AWS, GCP, Gradio, Streamlit, Databricks, Git, Linux

EXPERIENCE

Associate Analyst (AI/ML) | Highspring (On-site, Google)
Jun 2025 – Present

Gurugram | Python, Scikit-learn, LangChain, LLM Evaluation, SQL, Knowledge Graph, GCP

- Engineered a **hybrid KG + ML + LLM Wikipedia analysis pipeline** (scraping, NLP extraction, Gemini 3.5 Flash) across **10 locales and 5 verticals, covering ~5M entities**. Compressed a **months-long manual content-gap process to hours (~40% cost reduction)**.
- Designed a **LightGBM content-richness classifier** (Good/Average/Poor) to predict Comprehensiveness and Readability, trained on **400+ labelled articles** with per-vertical models and locale-calibrated thresholds. Benchmarked Wikipedia depth against KG entries and found **29.3% of entities were richer on Wikipedia**, quantifying the business case for acquisition.
- Built an **LLM Evaluation framework scoring 1,000+ LLM responses/day at ~85% F1 score** across correctness, consistency, relevance & completeness, **automating ~90% of manual QA effort** and driving Gemini model/prompt iteration decisions.
- Validated **70+ upstream data-provider pipelines across 1.4M+ entities**; found **17 providers drove ~63.8% of impact**, giving a clear prioritization signal for provider investment.

Data Science Intern | Haqdarshak
Feb 2025 – Jun 2025

Delhi | Python, LangChain, FastAPI, AWS

- Built and shipped a **video-generation pipeline that produced 350+ regional-language videos**, **cutting manual production effort by 70%**.
- Built a **scalable REST API** to automate script generation and database updates, streamlining content workflows for the team.
- Partnered with cross-functional teams to align solutions with business goals, contributing to a **40% increase in regional user engagement**.

Backend Engineer Intern | Verbaflo.AI
Sep 2024 – Dec 2024

Gurugram | Python, FastAPI, MongoDB, Docker

- Built **audit logging** for database operations and system tracking to support compliance and debugging.
- Engineered **JWT-based OAuth 2.0 authentication** and end-to-end CRM workflows from scratch, enabling secure role-based access.
- Developed **10+ REST APIs** for a Property Management System, **improving data retrieval efficiency by 35%**.

Machine Learning Intern | DraftAi
Jul 2024 – Aug 2024

Delhi | Python, LangChain, Flask

- Applied advanced **RAG techniques** to legal document generation, achieving a **62.5% improvement in legal citation accuracy over GPT-4o**.
- **Fine-tuned LLMs** for domain-specific legal workflows, **reducing hallucinations** in generated legal documents.
- Scaled the pipeline to process **1,000+ legal pages at 90%+ relevance**, overcoming context-window limits through chunking and retrieval design.

PROJECTS

Secure RAG with Role-Based Access Control (RBAC) | *GitHub*

Python, FastAPI, ChromaDB, Ollama, Streamlit

- Built an **end-to-end RAG assistant** (FastAPI + ChromaDB + Streamlit) **enforcing role-based access control at the vector-store level** — junior users cannot retrieve C-suite documents, **even via prompt injection**.
- Designed a **4-tier clearance + multi-tenant access model** with **JWT AuthN/AuthZ, prompt-injection guardrails, and full audit logging** — validated with an **automated test suite proving cross-tenant and cross-clearance isolation**.

ACHIEVEMENTS & LEADERSHIP

- **Certificate of Appreciation, Highspring** — recognized for the **Wikipedia content-gap analysis project** (~5M entities, 10 locales).
- **Top 10 of 150+ teams**, Devcation 2023 Hackathon, IGDTUW Delhi.
- **ML Core Team Member**, Intel oneAPI Students Club, GGSIPU (Aug 2023 – Jul 2024).
- **Open Source:** Contributor, GSSoC 2023 & Hacktoberfest 2023; **Mentor, GSSoC 2024**.

EDUCATION

Guru Gobind Singh Indraprastha University, Delhi
Aug 2021 – May 2025

B.Tech Artificial Intelligence and Data Science — CGPA: 8.59